التكنولوجيا التطبيقية
APPLIED TECHNOLOGY
وزارة التربية والتعليم والتعليم الفني

جمهورية مصر العربية
وزارة التربية والتعليم والتعليم الفني
وحدة تشغيل وإدارة مدارس التكنولوجيا التطبيقية

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Student guide

Physics

Year 1

(STEM)

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Unit three
Motion and force

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Dear student, By the end of this lessons you should have the following skills and knowledges

Apply the relation between the force, mass, and acceleration
Explain the action and reaction phenomenon
Explain the scientific meaning of Work.
Deduce that work is a non-vector quantity.
Apply the relation between work and force
Deduce units of Energy.
Compare between the kinetic energy and the potential energy.
Identify the mathematical relationship for both kinetic energy and potential energy.
Identify that potential energy is the work done
Differentiate between the concept of energy and power
Deduce units of power.
Apply the relation between the energy and power

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Lesson 1
Newton's laws of motion

Introduction

The previous chapter dealt with the description of linear motion in one direction (straight line) by studying the concepts of displacement, velocity and acceleration without getting into the reasons beyond.

In this chapter, we will discuss why objects move with acceleration as a result of an external effect (force), this effect can change the velocity of the object. The relationship between force and acceleration will be discussed by studying Newton's laws of motion that are considered as basic laws in physics.

The concept of force F

(An external effect that affects the object, it changes or attempts to change its state or direction).

OR

(It is an action which produces or tends to produce a change in a body's state of rest or motion).

Force (F) is a derived vector quantity, its measuring unit Newton (N) and measured by a spring balance.

Newton's first law

It is known that a rolling object on the floor would move for a certain distance, slowing down, and then stops due to presence of friction forces that slow it down till stop. If these forces do not exist, the object would keep moving with the same velocity in the same direction and would not stop. This principle is known as Newton's First Law of motion.

Statement of Newton's first law

"A static object keeps its state of rest, and a moving object keeps its state of motion at uniform velocity in a straight line unless acted upon by a resultant force."

That is mean, if the resultant force acting on the object is zero, the velocity of the object does not change, therefore the acceleration by which the object is moving equals zero.

The mathematical formula of Newton's first law

$$\sum F = \text{zero}$$

There are many examples, for example, the book of Physics (the object) which is placed on a horizontal table (contact case) under effect of two forces, one of them is weight of the book affects downward and the other the force by which the table affects the book. It is a normal force affects the object by the surface when they in contact case.

The relationship between the force of friction and the distance covered

1- If the force of friction increases, the distance covered decreases
2- If the force of friction decreases, the distance covered increases
3- If there is no frictional force, the object will continue to move in a straight line at a constant velocity

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Inertia

Newton's First Law is related to the concept of **inertia**. Accordingly, it is called (Law of Inertia)

(It is the tendency of an object to keep either its state of rest or state of motion at its original velocity uniformly in a straight line)

This mean that, the tendency of a static object to remain in a state of rest and the tendency of a moving object to continue moving at its original velocity in a straight line.

This means that objects resist changing its static or dynamic state.

From daily observations that illustrate the concept of inertia

1- The driver of the car rushes forward when stopping suddenly (when using the brakes suddenly) or when the car hits an obstacle. (therefore Seat belt should be fastened on driving)
2- Passengers suddenly rush back when the vehicle is suddenly moving
3- Passengers rush towards the car door when the car is traveling in a circular curve

Notes

It is observed that the possibility of stopping objects moving under the effect of inertia depends on the mass and speed of these objects, as:

1- A big truck is hard to stop while a small bike is easier to stop assuming they are moving at the same velocity.
2- It is difficult to stop the car if its velocity is high, while it is easy to stop if its velocity is low.

Some concepts related to Force

1- linear momentum

Linear momentum (P_L) is a derived vector quantity, its measuring unit (kg.m/s)

The mathematical formula of linear momentum

Momentum = mass × velocity

$$P_L = m\,v$$

2- Mass and Weight

Mass (m)

(Resistance of the object to change its Kinematic state)

Weight (F_g)

(It is the force of gravity acting on the object and its direction is towards the Earth's center)

Comparison between mass and weight

Quantity	Symbol	Mathematical Formula	Measuring Unit	Notes
Weight	F_g	$F_g = m\,g$	Newton (N)	1- It changes from one place to another according to the acceleration due to gravity 2- measured by the spring balance
Mass	m	m	Kilogram (kg)	1- It does not change from place to place

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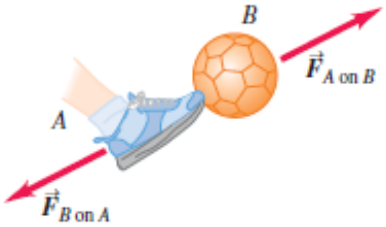
				2- measured by a beam balance or digital balance

Newton's third law

Newton formulated an explanation for the above mentioned situations through his third law of motion that studies the nature of forces acting on objects. He noticed that forces act in pairs of equal magnitude and opposite directions.

From daily observations



1- When the player kick a football, the forward force that your foot exerts on the ball launches it into its trajectory, but you also feel the force the ball exerts back on your foot.
2- The rifle bounces back when the bullet goes out from it.
3- The idea of operation of the rocket based on Newton's third law, where a huge mass of flaming gases rushes from the bottom of the rocket, and the reaction of the rocket is to rush upwards.
4- Consider the motion of a car on the way to school. A car is equipped with wheels that spin. As the wheels spin, they grip the road and push the road backwards. Since forces result from mutual interactions, the road must also be pushing the wheels forward. The size of the force on the road equals the size of the force on the wheels (or car); the direction of the force on the road (backwards) is opposite the direction of the force on the wheels (forwards). For every action, there is an equal (in size) and opposite (in direction) reaction. Action-reaction force pairs make it possible for cars to move along a roadway surface.

Newton's Third Law of motion statement

(When an object A acts on another object B by a force, the second object B acts on object A by a force equal in magnitude and opposite in direction)

OR

(For every action there is a reaction equal in magnitude and opposite in direction).

Notes

1- The statement of **Newton's third law means that** in every interaction, there is a pair of forces acting on the two interacting objects. The size of the forces on the first object equals the size of the force on the second object. The direction of the force on the first object is opposite to the direction of the force on the second object.
Forces always come in pairs - equal and opposite action-reaction force pairs

No single force exists in the universe. Action and reaction are paired; originate and vanish together.

i.e. Action and reaction are of the same type; if the action is a gravitational force, reaction is a gravitational force, as well.

2- Why cannot say that resultant of the action and reaction are at equilibrium?

For two forces to be at equilibrium, they must be equal, opposite, having one line of action and act on the same body. All these conditions except the last one may be applied on action and reaction; since they may act on different bodies

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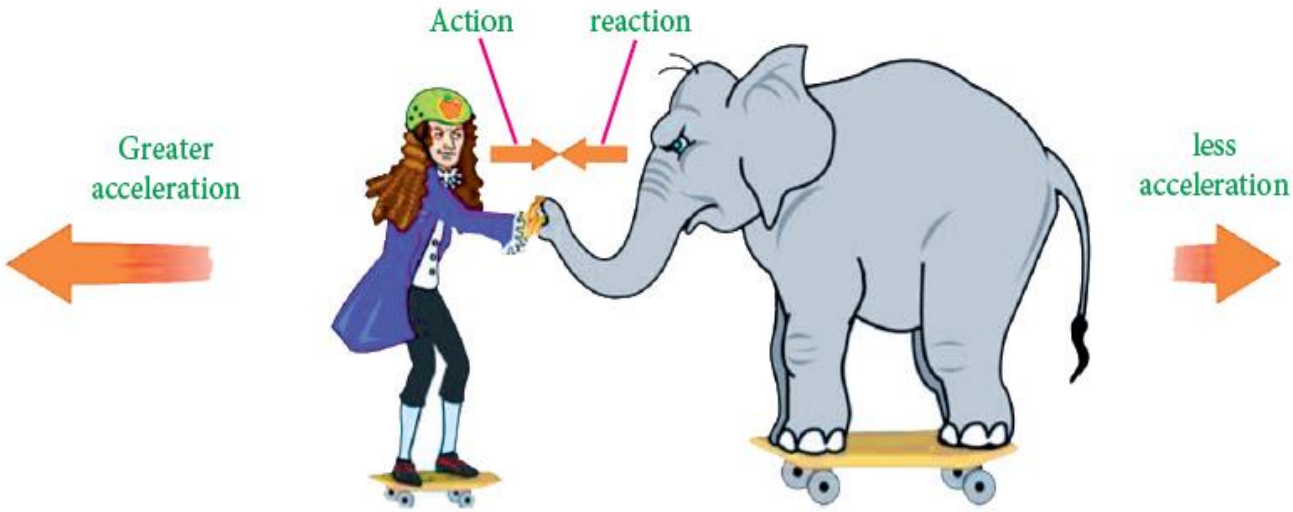
The mathematical formula

$$F_{A\ on\ B} = -F_{B\ on\ A}$$

<u>Exercise</u>

Study the following figure, then answer the questions below:

1- What is the relation between the force acting on the elephant and that on the man?
2- Why are not action on the elephant and reaction on the man at equilibrium?
3- If the elephant's mass is **6** times heavier than the man's mass, calculate the acceleration by which the elephant moves giving that the man moves at an acceleration $2m/s^2$. Why is the elephant acceleration negative?



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Lesson 2
Newton's second law

Introduction

We have learnt that when no resultant force affects the object, it does not move at acceleration. Consequently, when a resultant force acts on the object ($\Sigma F \neq 0$), it moves at acceleration. Accordingly, its velocity changes and it acquires acceleration ($a \neq 0$).Newton defined the factors affecting such acceleration through his second law

Newton's second law statement

(The resultant force acting on an object is equal to the time rate of change in the momentum of that body)

OR

(The resultant force acting on an object is equal to the product of the object's mass and its acceleration)

Mathematical formula of Newton's second law

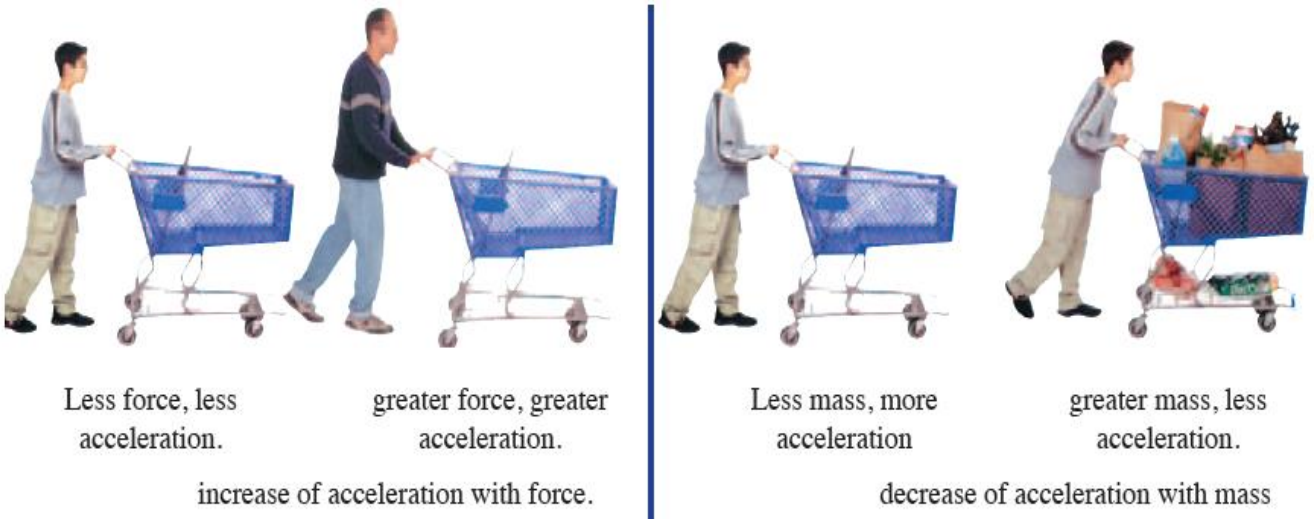
$$F = \frac{\Delta P_L}{\Delta t} = m\,a$$

Notes

"When a resultant force affects an object, the object acquires an acceleration which is directly proportional to the resultant force and inversely proportional to the object mass."

$$a = \frac{F}{m}$$

1- The acceleration is inversely proportional to the mass of the object at constant the resultant force.
2- The acceleration is directly proportional to the resultant force acting on the object at constant the mass of the object.



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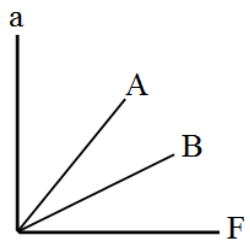
Exercise

1- The opposite graph represents the relationship between acceleration (a) acquired by the object and the resultant force (F) acting on the object of two objects A and B. Which of the two objects has the greater mass? Why?



2- A force of F acts on an object and accelerates it at certain acceleration (a).When another force acts on the same object and accelerates it at acceleration (5a). What do you conclude about the two forces acting on the object?

Through Newton's Second Law, we can infer that

1- Moving or stopping a heavy object as a train is much difficult than a lighter body as a bicycle.
2- Any object that acquires acceleration must exist force acting on it, and if an object falls, it moves with the free fall acceleration, which means that it is affected by a force known as the force of gravity.

Solved examples

Example (1)

Two equal forces act on two objects: the first has a mass of 10 kg, and it moves with an acceleration of 5 m/s^2, and the second has a mass of 25 kg. What is the acceleration of the second object?

The solution

$$\frac{m_1}{m_2} = \frac{a_1}{a_2}$$
$$\frac{10}{25} = \frac{a_2}{5}$$
$$a_2 = 2m/s^2$$

Example (2)

What is the force required for a car of mass 1.6×10^3 kg to move with a positive acceleration of 2m/s^2?

The solution

$$\because F = m\,a$$
$$\therefore F = (1.6\times10^3)(2)$$
$$\therefore F = 3.2\times10^3 \text{ Kg}$$

Example (3)

A car of mass 5000 kg started its motion from rest under effect a force of 4000 N, calculate the acceleration.

The solution

$$F = m\,a$$
$$a = \frac{F}{m} = \frac{4000}{5000} = 0.8\ m/s^2$$

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Lesson 3
Work

Introduction

In nature, energy exists in different forms such as thermal (heat) energy, chemical energy, mechanical energy, etc. this energy can be converted from one form into another. What is energy? What is its relation to the work done?

The term work may be used in our daily experience to express a task that captured somebody's interest and is involved in the most. This task could be mental as doing homework, physical as visiting a patient, or even just being at office. Physicists use the term work to express something really different

Work:

When a force acts upon an object to cause a displacement of the object, it is said that work was done upon the object.

There are three key ingredients to work - force, displacement, and cause. In order for a force to qualify as having done work on an object, there must be a displacement and the force must cause the displacement.

There are several good examples of work that can be observed in everyday life:

1- A Caw pulling a plow through the field.
2- A father pushing a grocery cart down the aisle of a grocery store.
3- A freshman lifting a backpack full of books upon her shoulder.
4- A student lifting a barbell above his head, an Olympian launching the shot-put.
5- A teacher applies a force to a wall and becomes exhausted.
6- A book falls off a table and free falls to the ground. etc.

In each case described here there is a force exerted upon an object to cause that object to be displaced.

If the object does not move, no work is done regardless how huge the exerted force is.

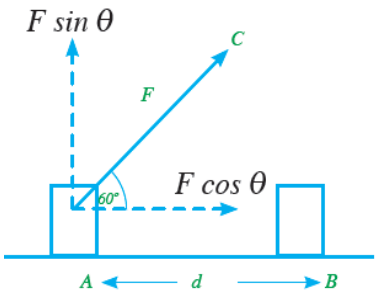
Accordingly**, there are two conditions for work to be done:**

1- A certain force acts on the object.
2- The object should move through a certain displacement in the direction of the force.

Mathematical formula of Work

If work done (W) by a certain force (F) on an object to move it through a displacement (d) along the line of the force action can be found by the relation:

$$W = F\,d$$



If the direction of the force (F) is inclined at an angle (θ) to the direction of displacement (d) as shown in figure, work done can be found by the relation:

$$W = F\,d\cos\theta$$

Measuring unit of work:

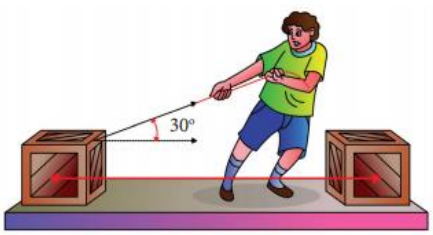
Work is a derived scalar quantity and its measuring unit (kg .m^2/ s^2) (Joule)
The Joule is the unit of work.
1 Joule = 1 Newton * 1 meter
1 J = 1 N * m
The Joule: is the work done by a force of one Newton to move an object through a displacement of one meter in the direction of the force.

What do you think?

1- When a rocket accelerates through space. Does the rocket do work?
2- When a waiter carries a tray full of meals above his head by one arm straight across the room at constant speed. Does the waiter do work?

Solved examples

Example 1:
A box is pulled with a force of 25 N to produce a displacement of 15 m. If the angle between the force and displacement is 30°, find the work done by the force.



Solution:

$$W = Fd \cos \theta$$
$$W = 25 \times 15 \times \cos 30 = 324.76 \text{ J}$$

Example 2:
If an object of mass 2 kg is thrown up from the ground reaches a height of 5 m and falls back to the Earth (neglect the air resistance). Calculate
a- The work done by gravity when the object reaches 5 m height
b- The work done by gravity when the object comes back to Earth
c- Total work done by gravity both in upward and downward motion and mention the physical significance of the result.

Solution:
When the object goes up, the displacement points in the upward direction whereas the gravitational force acting on the object points in downward direction, then the angle between gravitational force and displacement of the object is 180°
The work done by gravitational force in the upward motion

$$F = mg$$
$$W = F.d \cos \theta = mgd \cos \theta$$
$$W_{up} = 2 \times 10 \times 5 \cos (180) = -100 \text{ J}$$

When the object falls back, both the gravitational force and displacement of the object are in the same direction. This implies that the angle between gravitational force and displacement of the object is 0°.

$$W = F.d \cos \theta$$
$$W_{down} = 2 \times 10 \times 5 \cos (0) = 100 \text{ J}$$

The total work done by gravity in the entire trip (upward and downward motion)

$$W_{total} = W_{up} + W_{down} = -100 + 100 = 0$$

Example 3:

Calculate the work done by this girl who is carrying a bucket of mass (300 g) to move it through a displacement of (10 m) in the horizontal direction. Then, calculate the work done by the boy to lift a bucket of the same mass (10 cm) in the vertical direction. ($g = 10\ m/s^2$)

The solution

Work done by the girl:

Since the force exerted by the girl is perpendicular to displacement, work done equals zero.

Work done by the boy:

$$F = mg = 300\times10^{-3} \times10= 3N$$

Since force and displacement are in the same direction, $(\theta) = 0$

$$W = Fd \cos \theta = (3)\ (10\times10^{-2})\ (\cos 0) = 0.3\ J$$

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Lesson 4
Energy and some energy forms

Introduction

We use energy in the form of firewood, fossil fuels and electricity to make life comfortable and convenient.

At home we use electricity for our lights and fans, air-conditioner, water heater and room heaters, oven, microwave, washing machine and drier etc.

We use petrol, diesel, natural gas for our cars, buses, autos etc. Large amount of energy is consumed in agriculture and industry.

In offices we use energy to run air conditioners, fans, lights, computers, copying machines etc.

Energy is essential to life and all living organisms. The sun, directly or indirectly, is the source of all the energy available on Earth. Our energy choices and decisions impact Earth's natural systems in ways we may not be aware of, so it is essential that we choose our energy sources carefully.

The true cost of energy is more than just dollars and cents; there are important economic, political, and social factors and consequences to consider as well.

Energy:

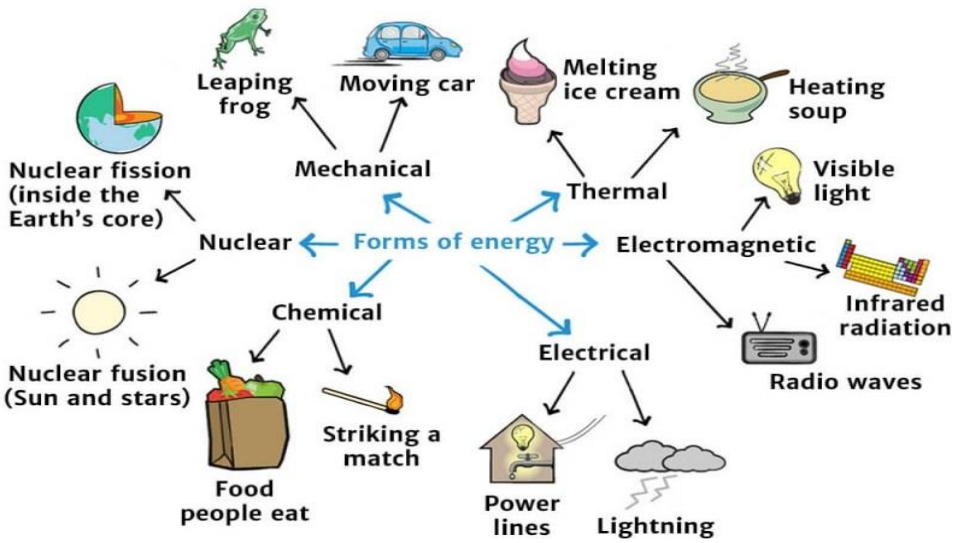
(It is the ability to do work)

The energy means that if a certain body is able to do work, this body possesses energy. Simply, the energy of a body is its capacity to do work. Because of this, the unit of energy is the same as the unit of work which is the Joule.

Energy is a derived scalar quantity and its measuring unit)(kg .m^2/ s^2) (Joule)

There are many different forms of energy, including

- **Heat (thermal)**
- **Light**
- **Mechanical**
- **Electrical**
- **Chemical**
- **Gravitational**
- **Nuclear**
- **Electromagnetic**



In the following section we are going to discuss two forms of energy which are kinetic energy and potential energy

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First: Kinetic Energy (KE)

When an object is acted upon by a force and the object starts to move by the impact of this force, we can say that this object has energy called kinetic energy (*K.E*).

Kinetic Energy The energy possessed by the object due to its motion.

Generally, kinetic energy of an object moving at velocity (v) and its mass (m) is calculated by the relation:

$$K.E = \frac{1}{2} m v^2$$

From the previous relation, we conclude that the kinetic energy is directly proportional to the object mass and the square of the object speed.

Note

The work done (the energy required to move the object) equals the kinetic energy. The work done is directly proportional to the square of the object speed. If we'd like to stop a car moving at velocity (60 km/h) by applying the brakes, the car would slide for a distance before stopping. This distance is four times longer than that needed if the car moves at velocity (30 km/h).

Solved example

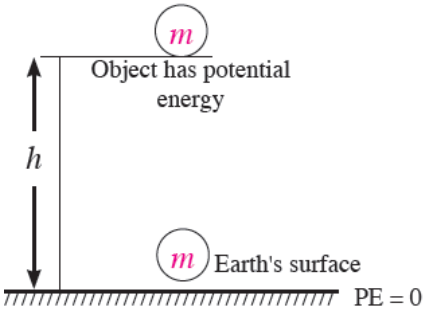
Find the kinetic energy of a car of mass (2000 kg) traveling at (72 km/h).

The solution

$$K.E = \frac{1}{2} m v^2 = \frac{1}{2} \times 2000 \times \left(72 \times \frac{1000}{60 \times 60}\right) = 4 \times 10^5 \text{ J}$$

Second: Potential Energy (P.E)

Energy can be stored in objects because of their new positions. This energy is called potential energy (PE).



Potential Energy the energy possessed by the object due to its position or state

For example,

1- Compression or elongation of a spring makes its particles have new positions and hence extra potential energy known as elastic potential energy. The spring then does work to release this energy, restoring its original position.

2- When an object is raised higher above the ground, it acquires potential energy known as gravitational potential energy. This energy is related to the object position relative to the surface of Earth.

Generally, **Potential** energy of an object of its mass (m) at height (h) from the ground by the relation:

$$P.E = m g h$$

From the previous relation, we conclude that the potential energy is directly proportional to the object mass and the square height from the ground.

Note

The work done to lift the object to a certain point = potential energy of this object at that point.

Solved example

Calculate the work done to lift an object of mass (50 kg) to (2.2m) high above the ground. ($g = 10$ m/s^2)

The solution

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$$P.E = m\,g\,h = 50 \times 10 \times 2.2 = 1100\ J$$

Lesson 5
Power

Introduction

Energy is the capacity to do work and power is the rate of energy delivery.
Power is always dependent on work done, so if a person does work at different rates his power also differs at different times. This is where the concept of average-power comes into the picture.

Power

(The rate of doing work)
(It is the work done in unit time).
The SI unit of power is Watt (W) which is joules per second (J/s).
Sometimes the power of motor vehicles and other machines is given in terms of Horsepower (hp), which is approximately equal to 745.7 watts.

What is Average Power?

We can define average power as the total energy consumed divided by the total time taken. In simple language, we can say that average power is the average amount of work done or energy converted per unit of time.

The mathematical formula of Power:

$$\text{Power} = \frac{\text{Work}}{\text{time}} = \frac{W}{t}$$

Unit of Power:

The unit for standard metric work is the Joule and the standard metric unit for time is the second, so the standard metric unit for power is a Joule / second, defined as a Watt and abbreviated W.

Solved Example:

A 60-watt bulb is switched on 24 hours a day and there is another 60-watt bulb that is turned on for only 12 hours. Find the energy consumed by both the bulbs in one day.

Solution:

For the first 12 hours, both bulb A and B are turned ON, therefore,
Power = 60 + 60 = 120 watts **Energy = Power × Time** = 120 × 12 = 1.44 kWh
Now for the next 12 hours only bulb A would remain ON hence,
Power = 60 watts Energy = 60 × 12 = 0.72 kW h
In this scenario, the power consumed during the whole day varies as one bulb is turned ON for only 12 hours, so we have to calculate average power,
Average Power = Total energy consumed / Total time taken
Average Power = (1.44 + 0.72) / 24 = 0.092 kW

You must have noticed that electrical components and energy meters in homes use a unit called kWh (kilowatt-hour), which stands for 1 kilowatt of power has been consumed every 1 hour, where kWh is the unit of energy.

Unit four
Waves

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Dear student, By the end of this chapter you should have the following skills and knowledges:

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Lesson 1
Types of waves

Introduction

Concept of motion

The object is said to be at rest when the position of the object does not change with respect to the time. While the object that changes its position as the time passes, the object is said to be in motion.

Motion is the change in the position of the object with respect to a fixed point as the time passes**.**

Motion can be classified into two types:

1- **The Translational (Translatory) motion:** is the motion which is characterized by having a starting point and end point in a certain path. Such as motion in a straight line, projectile motion, and transportations.

2- **Periodic motion**: is motion of an object that regularly returns to a given position after a fixed time interval, such as vibrational (oscillatory) motion , wave motion and circular motion

Note

The circular motion is the motion that repeats itself over equal intervals of time. Such as motion of earth around its axis and around Sun

The vibrational motion is the motion that repeats itself over equal intervals of time. Such as motion of simple pendulum forth and back

Wave motion

Introduction

Many of us enjoy watching waves on the surface of water pushing a fishing float or a boat up and down, or even making waves by throwing a pebble in a pond or still water. Each pebble becomes a source of disturbance in the water, spreading waves as concentric circles.

You encounter waves all the time in your daily life. For example, you can see because of visible light waves and hear because of sound waves.

Examples

Waves on the surface of liquid, sound waves, waves on stretched strings

Electromagnetic radiations, such as **light, x-rays, radio waves**, **infrared** and **ultraviolet radiation**

Wave is a disturbance that propagates and transports energy in the direction of its propagation through medium or space.

The world is full of waves, the two main types being **mechanical waves** and **electromagnetic waves**.

Mechanical waves

It is the disturbance that propagates through a medium, such as sound waves, water waves. (Some physical medium is being disturbed; in our pebble example, elements of water are disturbed.)

Electromagnetic waves

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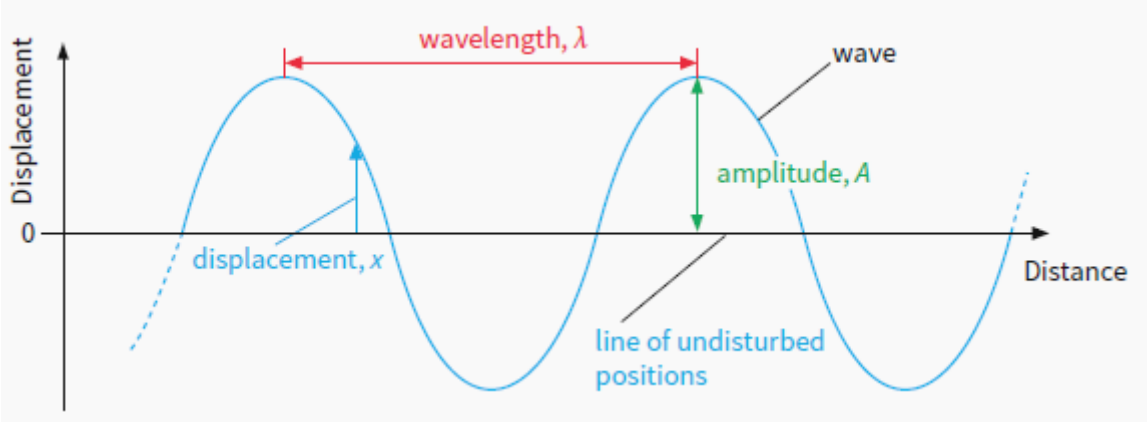
It is the disturbance that propagates through a medium or space, such as light waves, television signals, x- rays, radio waves and γ – rays (They do not require a medium to propagate)

Wave motion: energy is transferred over a distance, but matter is not.

Wave characteristics:



The following important definitions about waves and wave motion:

1) The displacement x

The distance of a point on the wave from its undisturbed position or equilibrium position

2) The amplitude A

The maximum displacement of any point on the wave from its undisturbed position

Note

The amplitude of a wave on the sea is measured in units of distance. The greater the amplitude of the wave, the louder the sound or the rougher the sea

3) The wavelength λ

The distance from any point on a wave to the next exactly similar point (the two points are in phase) (e.g. crest to crest)

(The Greek letter lambda) The wavelength of a wave on the sea is measured in units of distance

4) The period (periodic time) T

(The time taken for one complete oscillation of a point in a wave)

OR

(It is the time taken for a point to move from one particular position and return to that same position, moving in the same direction)

It is measured in units of time

5) The frequency f

(The number of oscillations per unit time of a point in a wave)

The frequency of a wave is measured in units of reciprocal time (Hz)

Notes

$$\lambda = \frac{\text{total distance}}{\text{number of waves}}$$

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$$f = \frac{\text{number of waves}}{\text{total time}}$$

$$T = \frac{\text{total time}}{\text{number of waves}} = \frac{1}{f}$$

Types of waves

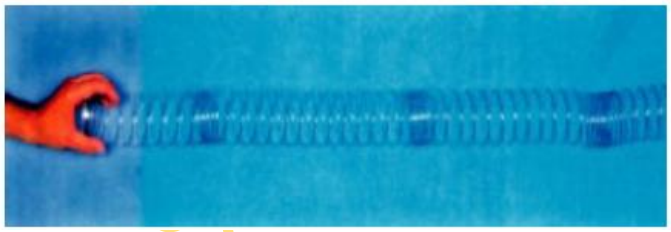
Mechanical Waves	Electromagnetic Waves
Mechanical waves need a material medium for their propagation.	Electromagnetic waves do not need any material medium for their propagation.
These waves are produced due to the vibrations of the particles of the medium.	They are caused due to change in electric and magnetic fields.
In this type of wave motion, the particles of the medium vibrate along or perpendicular to the direction of propagation of the wave	In this type of wave motion, the particles of the medium vibrate perpendicular to the direction of propagation of the wave
They have low speed	They have high speed
It is formed from Longitudinal and transverse waves	It is formed from transverse waves only
Ex. Water waves, Sound waves.	Ex. Radio waves, light, and X-rays

Applied

A Mechanical wave is a disturbance that travels through some material or substance called the **medium** for the wave. As the wave travels through the medium, the particles that make up the medium undergo displacements of various kinds, depending on the nature of the wave. There are two types of wave, **longitudinal and transverse**. Both can be demonstrated using a slinky spring.

Push the end of the spring back and forth; the segments of the spring become compressed and then stretched out, along the length of the spring. Wave pulses run along the spring. These are longitudinal waves.

Waggle the end of the slinky spring from side to side. The segments of the spring move from side to side as the wave travels along the spring. These are transverse waves. So the difference between longitudinal and transverse waves is as follows:



1) **In longitudinal waves**, the particles of the medium vibrate along (parallel to) the direction of the wave velocity. Such as sound waves and a wave travelling along a spring when it is pushed or pulled
2) **In transverse waves**, the particles of the medium vibrate perpendicular to (at right angles) the direction of the wave velocity. Such as waves formed on the surface of water and waves along a stretched string

<u>Wave energy</u>

It is important to realize that, for both types of mechanical wave, the particles that make up the material through which the wave is travelling do not move along – they only oscillate about a fixed point. It is energy that is transmitted by the wave. Each particle vibrates; as it does so, it pushes its neighbor, transferring energy to it. Then that particle pushes its neighbor, which pushes its neighbor. In this way, energy is transmitted from one particle to the next, to the next, and so on down the line.

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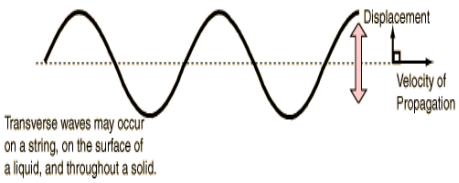
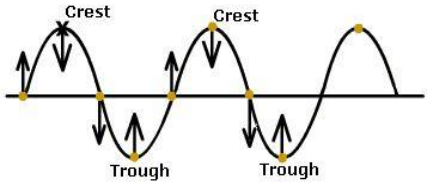
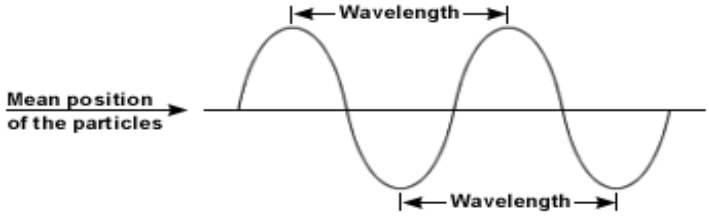
In transverse waves

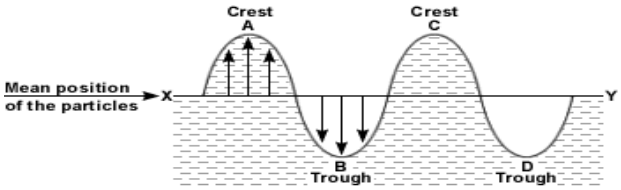
a) The particles of the medium vibrate perpendicular **(up and down)** to the direction of wave motion
b) The wave is propagated in the form of crests and troughs
c) **The wavelength** is the distance between two successive crests or two successive troughs.







Wavelength of Transverse Waves

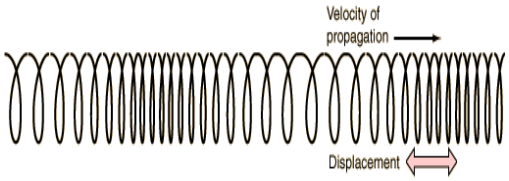


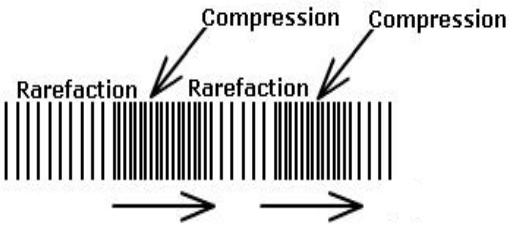
Crest and troughs of a transverse wave. Here, A and C are crests B and D are troughs

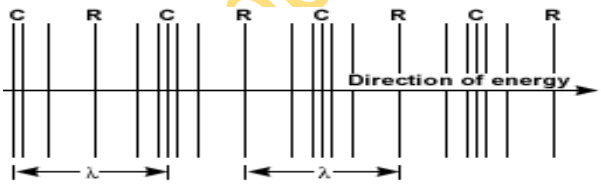
Note
Crest: it is the maximum displacement (amplitude) in the positive direction
Trough: it is the maximum displacement (amplitude) in the negative direction

In Longitudinal Waves

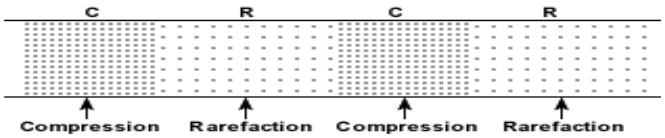
a) The particles of the medium vibrate along (parallel to) **(forth and back)** the direction of wave motion
b) The wave is propagated in the form of compressions and rarefactions
c) The wavelength is the distance between centers of two successive compressions or two successive rarefactions.







Wavelength of Longitudinal Waves



Compressions and rarefactions of a longitudinal wave. Regions marked C are compressions and regions marked R are rarefactions.

Note
Compression: areas of high pressure (the particles close to each other)
Rarefaction: areas of low pressure (the particles away from each other)

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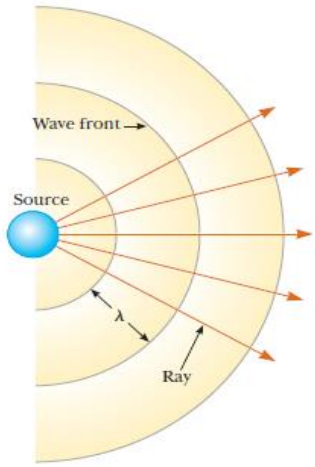
Lesson 3
Sound waves

Sound wave



Sound is a form of energy that travels in the form of a longitudinal mechanical wave in the form of compressions and rarefactions. Sound can travel through any medium, but it cannot travel through a vacuum. There is no sound in outer space**.**

Sound is produced by of vibration of sounding bodies. All sounding bodies vibrate but not all vibrating bodies cause sound. Because audible sounds

Humans are generally capable of hearing sounds between 20 Hertz and 20000 Hertz (20 kHz). Sounds with frequencies above the range of human hearing are called ultrasound. Sounds with frequencies below the range of human hearing are called infrasound.

Sound travels through air or any medium in all direction as spherical waves. Sound is a variation in pressure. A region of increased pressure on a sound wave is called **a compression**. A region of decreased pressure on a sound wave is called **a rarefaction**

Notes

For a gaseous medium

a) Speed of sound depends on the density of the medium that inversely proportional to the root of density of the medium $v \alpha \frac{1}{\sqrt{\rho}}$

b) The temperature of the medium also affects the speed of sound. As the temperature becomes higher, the density of the medium decreases, this make the sound waves travel faster.

Intensity

The term **intensity** has a meaning in physics. The intensity of a wave is defined as the rate of energy transmitted (i.e. power) per unit area at right angles to the wave velocity.

$$\text{intensity} = \frac{\text{power}}{\text{cross sectional area}}$$

Intensity is measured in watts per square meter **(W m^{-2})**.

Intensity and amplitude

The intensity **I** of a wave is related to its amplitude **A**. In fact, intensity is proportional to the square of the amplitude:

$$\text{Intensity } \alpha \text{ amplitude}^2 \ (I \ \alpha \ A^2)$$

Higher amplitude, higher wave intensity and vice versa

So, if one wave has twice the amplitude of another, it has four times the intensity. This means that it is carrying energy at four times the rate.

The intensity of a wave generally decreases as it travels along. There are two reasons for this:

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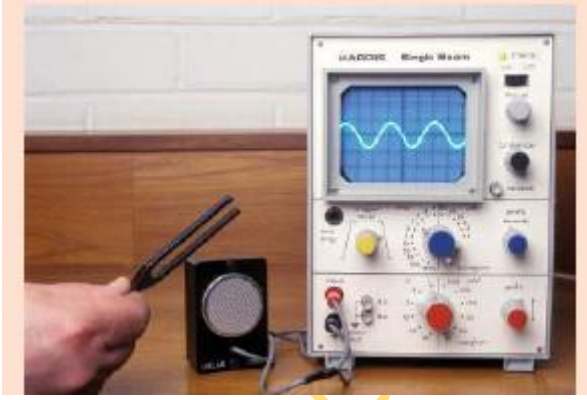
a) Spreading of the wave (As a wave spreads out, its amplitude decreases)
b) Absorption or Scattering of the wave

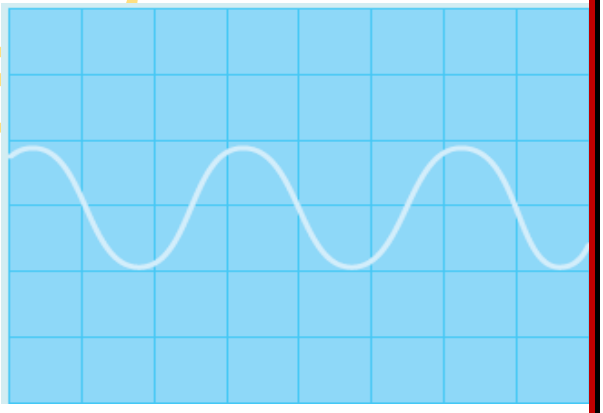
Measuring frequency



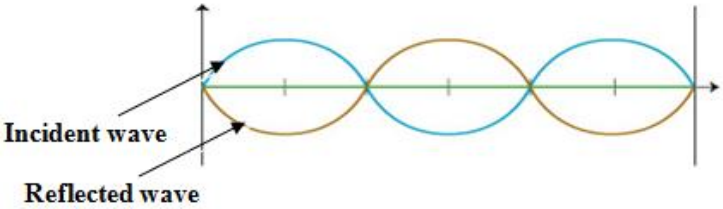
You can measure the frequency of sound waves using a cathode-ray oscilloscope (c.r.o.). The opposite figure shows how. A microphone is connected to the input of the c.r.o. Sound waves are captured by the microphone and converted into a varying voltage which has the same frequency as the sound waves. This voltage is displayed on the c.r.o. screen. It is best to think of a c.r.o. as a voltmeter which is capable of displaying a rapidly varying voltage. To do this, its spot moves across the screen at a steady speed, set by the time-base control. At the same time, the spot moves up and down according to the voltage of the input. Hence the display on the screen is a graph of the varying voltage, with time on the (horizontal) *x*-axis. If we know the horizontal scale, we can determine the period and hence the frequency of the sound wave

Exercise

The opposite Figure shows the trace on an oscilloscope screen when sound waves are detected by a microphone. The time-base is set at 1 ms/ div. Determine the frequency of the sound waves



Standing (stationary) Waves:



Consider a string, such as a guitar string, that is stretched between two clamps. Suppose we send a continuous sinusoidal wave of a certain frequency along the string, say, toward the right. When the wave reaches the right end, it reflects and begins to travel back to the left. That left-going wave then overlaps the wave that is still traveling to the right. When the left-going wave reaches the left end, it reflects again and the newly reflected wave begins to travel to the right, overlapping the left-going and right-going waves. In short, we very soon have many overlapping traveling waves, which interfere with one another. For certain frequencies, the interference produces a standing wave pattern (or **oscillation mode**) with nodes and large antinodes like those in Fig

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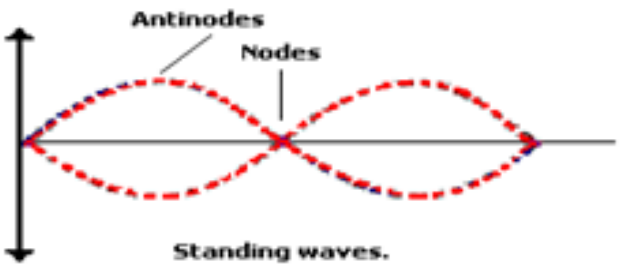
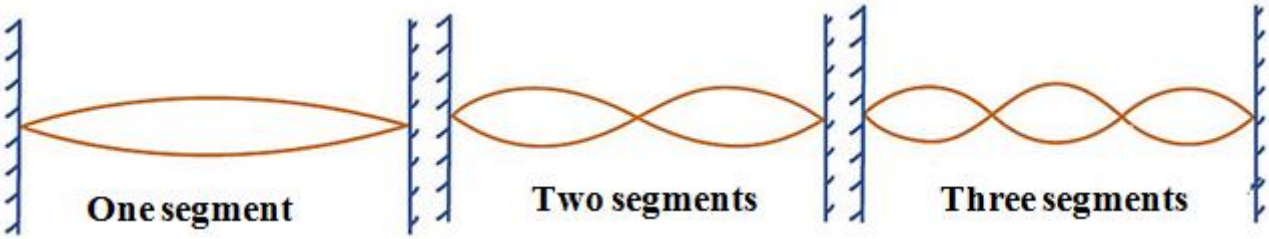
Standing Waves

The interference of two identical sinusoidal waves moving in opposite directions produces **standing waves**

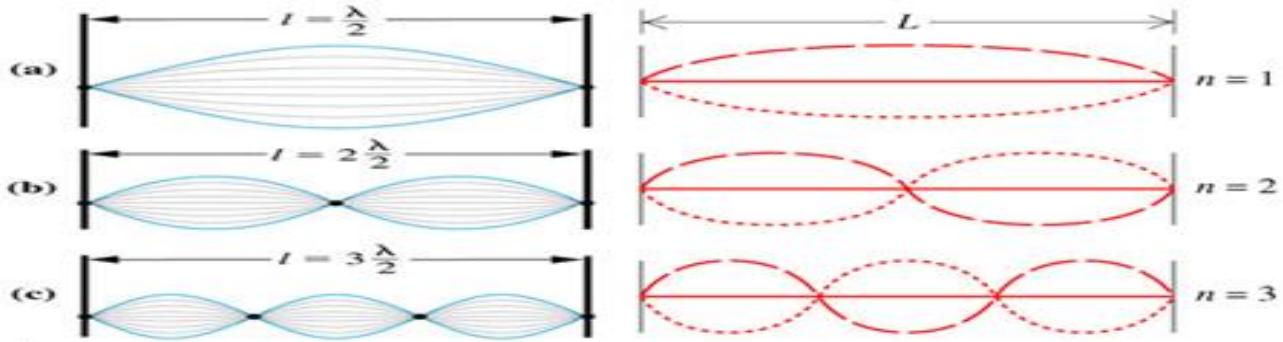
All **standing wave** patterns consist of nodes and antinodes where the nodes are produced at locations where amplitude vanishes and antinodes are produced at locations where the amplitude is a maximum



Variation of standing wave patterns



The relation to calculate the frequency of a tone produced from stretched string



For a stretched string of length L with fixed ends, the frequency of a tone is given by

$$f = \frac{n}{2L}\sqrt{\frac{F_T}{\mu}}$$

n: The number of segments (1, 2, 3,……)

L: The length of the string (measured in meter)

F_T: tension force (measured in Newton)

μ : mass per unit length (measured in kg/m)

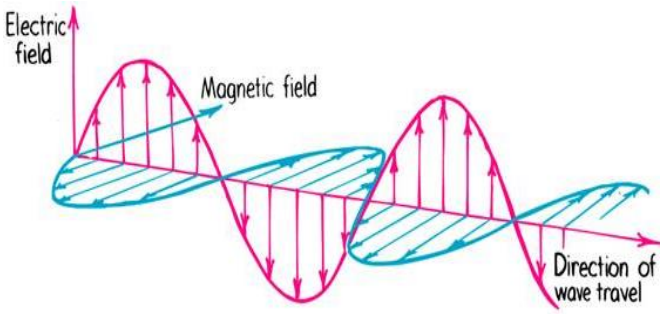
The oscillation mode corresponding to n = 1 is called the fundamental mode or the first harmonic; the mode corresponding to n = 2 is the second harmonic; and so on.

Lesson 4
Light

Introduction:

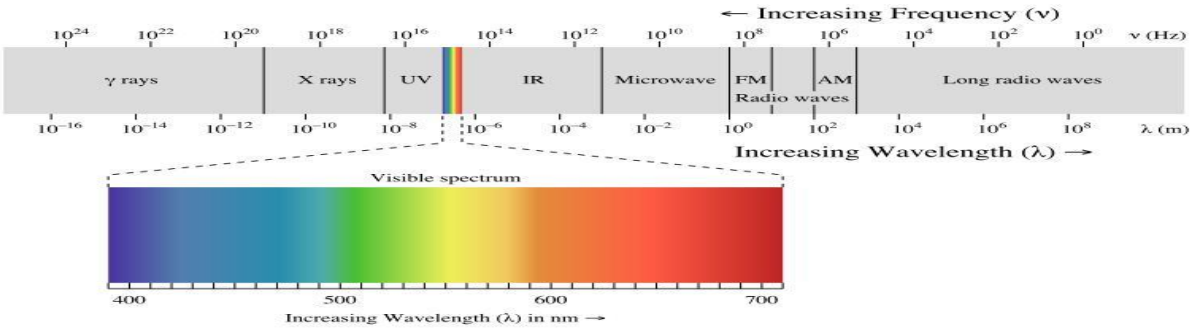
Electromagnetic waves:

They are vibrating electric and magnetic fields in phase, perpendicular to each other and perpendicular to the direction of their propagation.



By the end of the 19th century, several types of electromagnetic wave had been discovered:

1) **radio waves** – these were discovered by Heinrich Hertz when he was investigating electrical sparks
2) **infrared and ultraviolet waves** – these lie beyond either end of the visible spectrum
3) **X-rays** – these were discovered by Wilhelm Rontgen and were produced when a beam of electrons collided with a metal target such as tungsten
4) **γ-rays** – these were discovered by Henri Becquerel when he was investigating radioactive substances.



Properties of electromagnetic waves:

1) They are vibrating electric and magnetic fields that moves through space or any other medium.

2) They are transverse waves only.

3) They don't need medium to travel through it.

4) They propagate in space (as in air) at the same speed (c) = 3×10^{8} m/s while varying in frequency (and wavelength) and propagate in material media with a smaller velocity depending on the medium.

Light

Light is an Electromagnetic wave that occupies a small range in the electromagnetic spectrum

***Range of wavelength of light is between 400nm and 700nm.**
***light (visible light) consists of 7 colors (red-orange - yellow- green-blue –indigo - violet)**

The speed of light

James Clerk Maxwell showed that the speed *c* of electromagnetic radiation in a vacuum (free space) was independent of the frequency of the waves. In other words, all types of electromagnetic wave travel at the same speed in a vacuum. In the SI system of units, ***C*** has the approximate value for the speed of light in a vacuum
(Often used in calculations) is 3×10^8 m /s.
The wavelength λ and frequency *f* of the radiation are related by the equation:

$$c = f\lambda$$

When light travels from a vacuum into a material medium such as glass, its speed decreases but its frequency remains the same, and so we conclude that its wavelength must decrease. We often think of different forms of electromagnetic radiation as being characterized by their different wavelengths, but it is better to think of their different frequencies as being their fundamental characteristic, since their wavelengths depend on the medium through which they are travelling.

Light ray

To describe the directions in which light propagates, it's often convenient to represent a light wave by **rays** rather than by wave fronts. Rays were used to describe light long before its wave nature was firmly established. In a particle theory of light, rays are the paths of the particles. From the wave viewpoint a ray is an imaginary line along the direction of travel of the wave. The rays are always straight lines normal to the wave fronts. At a boundary surface between two materials, such as the surface of a glass plate in air, the wave speed and the direction of a ray may change, but the ray segments in the air and in the glass are straight lines

We briefly describe these different types of electromagnetic waves, in order of decreasing wavelengths.

Radio waves

Radio waves are produced by the accelerated motion of charges in conducting wires. They are used in radio and television communication systems. They are generally in the frequency range from 500 kHz to about 1000 MHz. The AM (amplitude modulated) band is from 530 kHz to 1710 kHz. Higher frequencies upto 54 MHz are used for *short wave* bands. TV waves range from 54 MHz to 890 MHz. The FM (frequency modulated) radio band extends from 88 MHz to 108 MHz. Cellular phones use radio waves to transmit voice communication in the ultrahigh frequency (UHF) band.

Microwaves

Microwaves (short-wavelength radio waves), with frequencies in the gigahertz (GHz) range, are produced by special vacuum tubes (called klystrons, magnetrons and Gunn diodes). Due to their short wavelengths, they are suitable for the radar systems used in aircraft navigation.

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Radar also provides the basis for the speed guns used to time fast balls, tennisserves, and automobiles. Microwave ovens are an interesting domestic application of these waves. In such ovens, the frequency of the microwaves is selected to match the resonant frequency of water molecules so that energy from the waves is transferred efficiently to the kinetic energy of the molecules. This raises the temperature of any food containing water.

Infrared waves

Infrared waves are produced by hot bodies and molecules. This band lies adjacent to the low-frequency or long-wave length end of the visible spectrum. Infrared waves are sometimes referred to as *heat waves*. This is because water molecules present in most materials readily absorb infrared waves (many other molecules, for example, CO2, NH3, also absorb infrared waves). After absorption, their thermal motion increases, that is, they heat up and heat their surroundings. Infrared lamps are used in physical therapy. Infrared radiation also plays an important role in maintaining the earth's warmth or average temperature through the greenhouse effect. Incoming visible light (which passes relatively easily through the atmosphere) is absorbed by the earth's surface and reradiated as infrared (longer wavelength) radiations. This radiation is trapped by greenhouse gases such as carbon dioxide and water vapor. Infrared detectors are used in Earth satellites, both for military purposes and to observe growth of crops. Electronic devices (for example semiconductor light emitting diodes) also emit infrared and are widely used in the remote switches of household electronic systems such as TV sets, video recorders and hi-fi systems.

Visible rays

It is the most familiar form of electromagnetic waves. It is the part of the spectrum that is detected by the human eye. It runs from about 4×10^{14} Hz to about 7×10^{14} Hz or a wavelength range of about 700 – 400 nm. Visible light emitted or reflected from objects around us provides us information about the world. Our eyes are sensitive to this range of wavelengths. Different animals are sensitive to different range of wavelengths. For example, snakes can detect infrared waves, and the 'visible' range of many insects extends well into the utraviolet.

Ultraviolet rays

It covers wavelengths ranging from about 4×10^{-7} m (400 nm) down to 6×10^{-10} m (0.6 nm). Ultraviolet (UV) radiation is produced by special lamps and very hot bodies. The sun is an important source of ultraviolet light. But fortunately, most of it is absorbed in the ozone layer in the atmosphere at an altitude of about 40 – 50 km. UV light in large quantities has harmful effects on humans. Exposure to UV radiation induces the production of more melanin, causing tanning of the skin. UV radiation is absorbed by ordinary glass. Hence, one cannot get tanned or sunburn through glass windows.

Welders wear special glass goggles or face masks with glass windows to protect their eyes from large amount of UV produced by welding arcs. Due to its shorter wavelengths, UV radiations can be focused into very narrow beams for high precision applications such as LASIK (*Laser assisted in situ keratom ileusis*) eye surgery. UV lamps are used to kill germs in water purifiers.

Ozone layer in the atmosphere plays a protective role, and hence its depletion by chlorofluorocarbons (CFCs) gas (such as Freon) is a matter of international concern.

X-rays

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Beyond the UV region of the electromagnetic spectrum lies the X-ray region. We are familiar with X-rays because of its medical applications. It covers wavelengths from about 10^{-8} m (10 nm) down to 10^{-13} m (10^{-4} nm). One common way to generate X-rays is to bombard a metal target by high energy electrons. X-rays are used as a diagnostic tool in medicine and as a treatment for certain forms of cancer. Because X-rays damage or destroy living tissues and organisms, care must be taken to avoid unnecessary or over exposure.

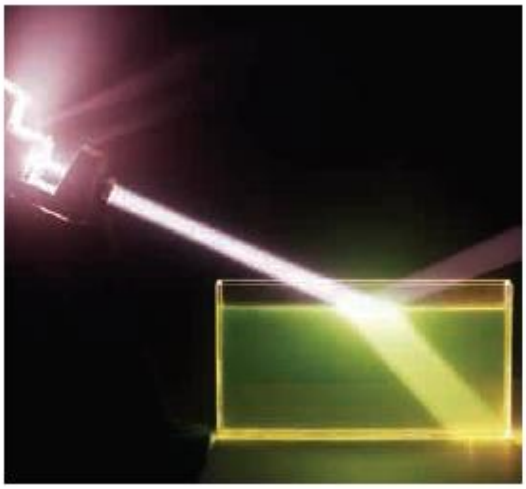
Gamma rays

They lie in the upper frequency range of the electromagnetic spectrum and have wavelengths of from about 10^{-10} m to less than 10^{-14} m. This high frequency radiation is produced in nuclear reactions and also emitted by radioactive nuclei. They are used in medicine to destroy cancer cells.

Type	Wavelength range	Production	Detection
Radio	> 0.1 m	Rapid acceleration and decelerations of electrons in aerials	Receiver's aerials
Microwave	0.1m to 1 mm	Klystron valve or magnetron valve	Point contact diodes
Infra-red	1mm to 700 nm	Vibration of atoms and molecules	Thermopiles Bolometer, Infrared photographic film
Light	700 nm to 400 nm	Electrons in atoms emit light when they move from one energy level to a lower energy level	The eye Photocells Photographic film
Ultraviolet	400 nm to 1nm	Inner shell electrons in atoms moving from one energy level to a lower level	Photocells Photographic film
X-rays	1nm to 10^{-3} nm	X-ray tubes or inner shell electrons	Photographic film Geiger tubes Ionisation chamber
Gamma rays	$<10^{-3}$ nm	Radioactive decay of the nucleus	-do-

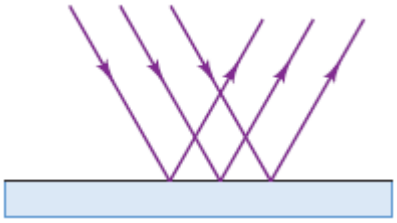
Lesson 5
Reflection and Refraction of light

Introduction:



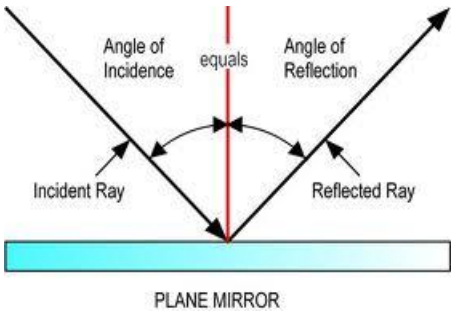
When a light wave strikes a smooth interface separating two transparent materials (such as air and glass or water and glass), the wave is in general partly reflected and partly refracted (transmitted) into the second material. Part of the light is **reflected** by the surface, forming a beam directed upward toward the right, traveling as if the original beam had bounced from the surface. The rest of the light travels through the surface and into the water, forming a beam directed downward to the right. Because light can travel through it, the water is said to be transparent; that is, we can see through it.

Reflection of light:



Reflection is defined as the bouncing back (return) of a ray of light into the same medium, when it strikes a surface (such as mirrors).

Reflection can occur with either transparent materials or opaque materials that do not transmit light



Laws of reflection:

1- Angle of incidence = angle of reflection.

2-The incident ray , the reflected ray & the normal to the surface of reflection at the point of incidence , all lie in one plane perpendicular to the plane of the reflecting surface.

Refraction of light

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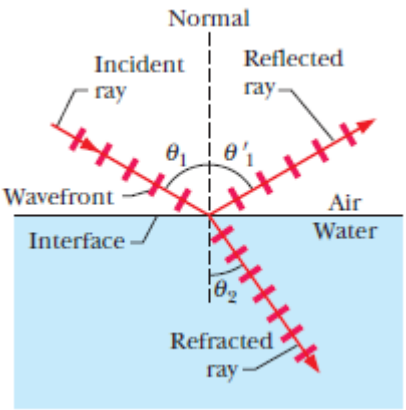
(It is the bending of a light wave when it passes from one medium to another at the surface separating the two media due to the change of speed of light in different media of different densities)



The travel of light through a surface (or interface) that separates two media is called **refraction,** and the light is said to be refracted. Unless an incident beam of light is perpendicular to the surface, refraction changes the light's direction of travel. For this reason, the beam is said to be "bent" by the refraction

The index of refraction of an optical material (the refractive index)

The refractive index (n): It is the ratio of the speed of light in vacuum to the speed in the material:

$$n = \frac{c}{v} = \frac{\sin\theta_{air}}{\sin\theta_{medium}}$$

Light always travels more slowly in a material than in vacuum, so the value of (n) in anything other than vacuum is always greater than unity. For vacuum, n = 1.

Wave speed and index of refraction

Keep in mind that the wave speed is *inversely* proportional to the index of refraction the greater the index of refraction in a material, the *slower* the wave speed in that material

Laws of refraction:

1) The ratio between the sin the angle of incidence to sin the angle of refraction is constant and is known as **the refractive index** between the two media (n_{12})

$$n_{12} = \frac{v_1}{v_2} = \frac{\sin\theta_1}{\sin\theta_2}$$

OR

$$n_1 \times \sin\theta_1 = n_2 \times \sin\theta_2 \qquad \textbf{(Snell's law)}$$

n_1: absolute refraction index of the first medium
n_2: absolute refraction index of the second medium
v_1: speed of light in the first medium
v_2: speed of light in the second medium
θ_1: angle of incidence in the first medium (The angle of incidence is measured with respect to the normal at the refractive boundary)
θ_2: angle of refraction in the second medium (The angle of refraction is measured with respect to the normal at the refractive boundary)

2) The incident ray, refracted ray and the normal to the separating surface at the point of incidence lie in the same plane

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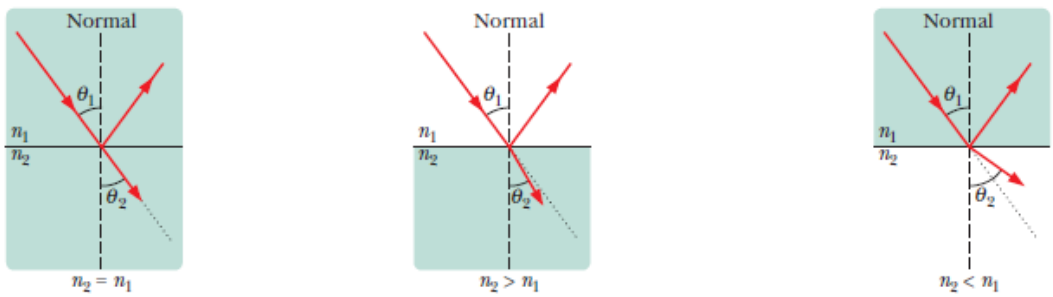
Notes

$$n_{12} = \frac{v_1}{v_2} = \frac{\sin\theta_1}{\sin\theta_2}$$

To compare the angle of refraction θ_2 with the angle of incidence θ_1, we can then see that the relative value of θ_2 depends on the relative values of n_2 and n_1:

1. If n_2 is equal to n_1, then θ_2 is equal to θ_1 and refraction does not bend the light beam, which continues in the ***undeflected direction***, as in figure below.
2. If n_2 is greater than n_1, then θ_2 is less than θ_1. In this case, refraction bends the light beam away from the undeflected direction and toward the normal, as in figure below
3. If n_2 is less than n_1, then θ_2 is greater than θ_1. In this case, refraction bends the light beam away from the undeflected direction and away from the normal, as in figure below.



(a) If the indexes match, there is no direction change.

(b) If the next index is greater, the ray is bent *toward* the normal.

(c) If the next index is less, the ray is bent *away from* the normal.

Refraction of light traveling from a medium with an index of refraction n_1 into a medium with an index of refraction n_2. (*a*) The beam does not bend when $n_2 = n_1$; the refracted light then travels in the *undeflected direction* (the dotted line), which is the same as the direction of the incident beam. The beam bends (*b*) toward the normal when $n_2 > n_1$ and (*c*) away from the normal when $n_2 < n_1$.

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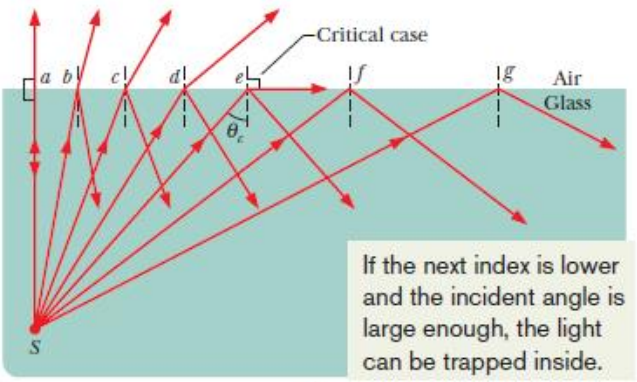
Lesson 6

Total internal reflection

The two figures below show rays of monochromatic light from a point source *S* in glass incident on the interface between the glass and air. For ray *a*, which is perpendicular to the interface, part of the light reflects at the interface and the rest travels through it with no change in direction. For rays b through *e*, which have progressively larger angles of incidence at the interface, there are also both reflection and refraction at the interface. As the angle of incidence increases, the angle of refraction increases; for ray *e* it is 90°, which means that the refracted ray points directly along the interface. The angle of incidence giving this situation is called the **critical angle** θ_c. For angles of incidence larger than θ_c, such as for rays *f* and *g*, there is no refracted ray and *all* the light is reflected; this effect is called **total internal reflection.**

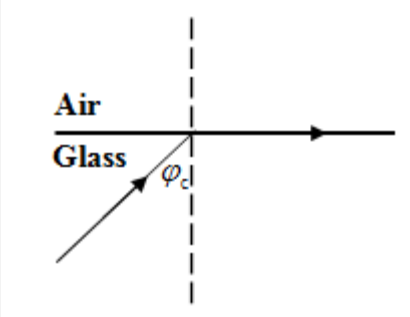


(a)

(b)

i.e.

1) When a ray of light passes from an optical medium of greater refractive index (n_1) (such as water or glass) to an optical medium of lower refractive index (n_2) (such as air). It refracts away from the normal by increasing the angle of incidence in the first medium, the angle of refraction in the second medium increase also.



2) When the angle of incidence reaches certain value, the corresponding angle of refraction reach 90° and hence the ray emerges along (parallel to) the separating surface. In this case the angle of incidence in the first medium is known as (**The critical angle** θ_c**)**

The critical angle

(It is the angle of incidence in the medium of greater refractive index (1st medium) that corresponds to an angle of refraction in the medium lower refractive index (2nd medium) equal to 90°)

To find the critical angle $\boldsymbol{\theta_c}$

$$n_1 \times \sin\theta_1 = n_2 \times \sin\theta_2$$

$\theta_1 = \theta_c$ (the critical angle) if $\theta_2 = 90^o$

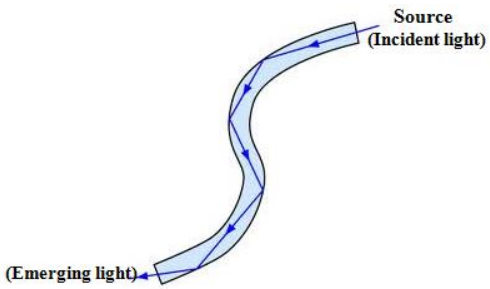
$$n_1 \times \sin\theta_c = n_2 \times \sin 90^o$$

$$\theta_c = \sin^{-1}\frac{n_2}{n_1} \quad \text{(critical angle).}$$

Application of total internal reflection:

The optical fibers:

Structure:



a) The optical fibers is a fine hollow tube of a transparent material (glass or plastic) of high refractive index (small critical angle)

b) When a ray of light enters the tube, it suffers successive reflections. (Since the angle of incidence is greater than the critical angle).Till it emerges from the other end of the tube without loss of energy.

Functions:

1) Fiber-optic devices have found a wide range of medical applications in instruments called *endoscopes*, which can be inserted directly into the bronchial tubes, the bladder, the colon, and other organs for direct visual examination.
2) A bundle of fibers can even be enclosed in a hypodermic needle for studying tissues and blood vessels far beneath the skin.
3) Fiber optics also have applications in communication systems, in which they are used to transmit a modulated laser beam. The rate at which information can be transmitted by a wave (light, radio, or whatever) is proportional to the frequency.

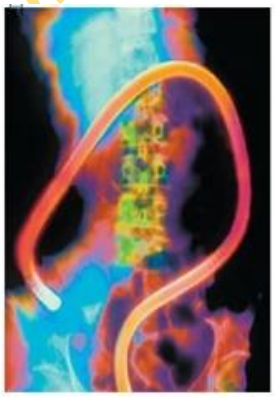


image of a patient's abdomen shows an endoscope winding through the colon.

To see qualitatively why this is so, consider modulating (modifying) the wave by chopping off some of the wave crests. Suppose each crest represents a binary digit, with a chopped-off crest representing a zero and an unmodified crest representing a one. The number of binary digits we can transmit per unit time is thus proportional to the frequency of the wave. Infrared and visible- light waves have much higher frequency than do radio waves, so a modulated laser beam can transmit an enormous amount of information through a single fiber-optic cable.

4) Fiber-optic cables play an important role in long-distance telephone, television, and Internet communication. The advantage of optical fibers is that they can be made thinner than conventional copper wire, so more fibers can be bundled together in a cable of a given diameter. Hence more distinct signals (for instance, different phone lines) can be sent over the same cable. Because fiber-optic cables are electrical insulators, they are immune to electrical interference from lightning and other sources, and they don't allow unwanted currents between source and receiver.

Note that:

Each fiber **coated with a thin layer of a material of lower refractive index**

a) To keep the light rays inside the core without escaping

b) To reflect any rays escaping from the core to keep the intensity of light without decrease.

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